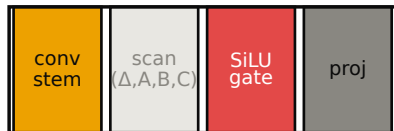


Step 1 -- Freeze the scan (necessity test)

conv stem scan (Δ, A, B, C) SiLU gate proj

BM3-frozen



Step 2 -- Remove gate (localise)

frozen – gate



Step 3 -- Add gate to S4D (sufficiency test)

S4D + gate
+ added



partial / full freeze vs S4D:
+12–18pp / +20–23pp (indep. noise)
→ input-indep. coefficients suffice

gate removal: +9 / +16 / +27pp
(resolved at –6 dB only)
→ gate need grows with noise

additive in place of gate $\approx$ no gate;
S4D: gate 0.30 vs additive 0.41
→ little gate-specific transfer

Conditional on this block, testbed and budget: input-independent scan coefficients suffice; the gate matters most in the deepest noise.